

Influence of Rice Husk Ash on Crystalline Phases and Mechanical Characteristics of Sustainable Bricks

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Abstract

This study investigates the use of rice husk ash (RHA) as a sustainable additive in ceramic brick production. RHA from the Chaco region of Argentina was analyzed using DTA/TG and XRD, revealing a decrease in the amorphous phase and an increase in crystalline phases, mainly quartz and albite, with higher RHA content. The incorporation of RHA increased porosity from 25% to 40%, while theoretical density remained stable. Mechanical tests showed a decrease in elastic modulus (72.76 GPa for 5% RHA to 71.74 GPa for 15% RHA) and modulus of rupture (MOR) (8.5 MPa for 5% RHA to 5 MPa for 30% RHA). Several models were analyzed to predict the elastic modulus, and the DEM model provided the best predictions, with a relative error of 1.43% for 15% RHA. The analysis of mechanical properties of porosity was conducted using various theoretical models, which allowed for a better understanding of the relationship between the porous structure of the bricks and their mechanical behavior. Theoretical models, such as the DEM, proved effective in describing how porosity influences mechanical properties, particularly elastic modulus and modulus of rupture. Despite the reduction in mechanical properties, bricks with up to 15% RHA are suitable for non-structural applications, offering an eco-friendly alternative to conventional bricks. This research provides a circular economy approach by transforming agricultural waste into valuable construction materials.

Keywords: Rice husk ash (RHA), Sintering, Porosity, Elastic modulus, Crystalline phases

INTRODUCTION

Rice cultivation is a significant agricultural activity that generates rice husk as a by-product. If not properly managed, this lignocellulosic material contributes to environmental pollution due to its long degradation time. According to data from the Ministry of Agriculture, Livestock, and Fisheries, Argentina produced 1.6 million tons of rice in 2023, with production increasing by 121% in the last decade, alongside a 16% rise in prices. In this context, rice husk has become a waste of considerable interest and concern, posing challenges for its effective management [1].

Rice milling industries often face the dilemma of how to manage this significant waste, as improper disposal methods, such as open burning, can lead to major environmental consequences. One alternative currently used by several rice producers is the use of rice husk as an energy source, burning it in boilers under controlled conditions for internal energy supply [2]. The resulting ash contains high concentrations of silica, which could pose risks to human and animal health as well as the environment. Addressing the treatment of this solid waste, which contributes to environmental pollution, is one of the most pressing issues today. There is a growing demand for effective waste management and disposal

methods that mitigate environmental harm.

The incorporation of lignocellulosic materials in the production of new products is an emerging trend, yielding products comparable to those made with conventional raw materials but in an environmentally friendly manner. Interest in the use of rice husk ash (RHA) has increased significantly in recent years across various industries, such as cement, glass, and ceramics, mainly due to its high SiO₂ content [3]. Numerous applications have been identified in the construction sector, where RHA is used as an additive in concrete production and in the manufacture of ceramic bricks, serving both structural and thermal/acoustic insulation purposes [4].

The ash generated by combustion is predominantly composed of silica (87%-97% by weight), with small amounts of inorganic salts containing elements such as potassium, sodium, and lithium. Its reactivity is influenced by several factors, including calcination temperature, chemical pretreatments, combustion time, heating rate, and the type of furnace used [3]. The properties of RHA can vary depending on combustion processes, maintaining a highly reactive amorphous structure up to approximately 600 °C, regardless of calcination time. However, at 700 °C and with prolonged exposure, the amorphous silica begins to rearrange into a less reactive crystalline state [5, 6]. Therefore, the ability to control the calcination process is crucial to optimizing RHA's performance in construction materials.

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The application of RHA in ceramic bricks offers significant advantages both environmentally and in terms of material performance. The incorporation of rice husk ash (RHA) in brick production not only helps reduce agro-industrial waste but also improves the final product's properties, providing a sustainable alternative to conventional building materials [7].

One of the main advantages of using RHA is its ability to improve the thermal and acoustic properties of bricks. Due to its porous structure, RHA contributes to higher thermal insulation capacity, making buildings more energy-efficient and reducing the need for heating or cooling. Additionally, its porous characteristics provide good acoustic insulation, enhancing comfort inside buildings [8, 9]. This increase in porosity could also reduce the density of the bricks, making them lighter and easier to handle, although the amount used should be controlled to avoid compromising their mechanical strength.

From an environmental perspective, the use of RHA would significantly reduce the carbon footprint in the construction industry. By partially replacing conventional materials such as clay or cement with an agricultural by-product, the extraction of natural resources would be reduced, and energy consumption in brick production would decrease. This would lead to a reduction in CO₂ emissions associated with the manufacturing of traditional bricks.

Additionally, the circular economy would be another key benefit. By recycling RHA, value is given to a by-product of the rice industry that would otherwise be disposed of improperly, which would also help reduce pollution from the open burning of rice husk. This approach not only mitigates environmental impact but also offers a source of local, affordable raw material, especially in rice-producing regions such as the Argentina Chaco.

In conclusion, the application of RHA in ceramic brick production would not only offer a sustainable and cost-effective alternative to conventional construction methods but also promote resource-efficient use and contribute to the development of a more eco-friendly and responsible construction model, in line with the principles of the circular economy.

In this study, ceramic bricks based on rice husk ash (RHA) and clays from the Chaco region were produced with varying ash contents (5%, 15%, and 30% by weight). The effect of the phase composition of the bricks (i.e., different ash-to-clay ratios) on textural properties such as density and porosity and mechanical properties, including dynamic Young's modulus and modulus of rupture, was investigated. Predictive models, such as the DEM model, showed a strong fit with the experimental values, accurately describing the effect of porosity on the material's elastic properties.

The innovative use of rice husk ash not only addresses local waste management problems but also offers an environmentally sustainable solution for the construction industry. By transforming a by-product of rice milling into valuable construction materials, this research contributes to the principles of the circular economy, promoting

resource efficiency and reducing the environmental impacts associated with waste disposal.

The findings of this study provide a solid foundation for exploring the use of RHA in ceramic brick production, with the potential to improve material properties and contribute to more sustainable construction. This approach could offer new alternatives for brick manufacturing in various regional contexts, addressing the challenges associated with agricultural waste.

The implications of this research go beyond the Chaco region. The results highlight the potential of rice husk ash as a sustainable material for brick production in other regions facing similar agricultural waste challenges. As global interest in sustainable building materials continues to grow, the use of RHA could serve as a model for other countries seeking to valorize their agricultural by-products while improving the performance of their construction materials.

EXPERIMENTAL

The raw materials for the production of ceramic bricks included two types of soil extracted from the Chaco region (Sáenz Peña, Argentina): Black Soil (BS), which is rich in organic matter that increases porosity during firing, and Red Soil (RS), containing iron oxides, which provides good plasticity and strength after thermal treatment. [10-13]. Additionally, rice husk ash (RHA) was sourced from a factory in Las Palmas (Teko industry, Chaco, Argentina). The soils were conditioned by sieving through a N°. 60 mesh (250 µm). The RHA was produced on a laboratory scale via a controlled calcination process in an electric furnace, Nibias model HCF 70L, at 450 °C, with a heating rate of 5 °C/min for 150 min.

Several brick manufacturing series were conducted, starting with the traditional adobe method without the addition of rice husk ash (RHA) and incorporating varying RHA percentages (5%, 15%, and 30% by weight). The unfired bricks were designated as V0RHA, V-5RHA, V-15RHA, and V-30RHA.

The initial mixtures consisted of a 70 wt.% red soil and 30 wt.% black soil. This proportion was adjusted with the addition of rice husk ash (RHA) to ensure the cohesion and stability of the samples. The resulting pastes, with a moisture content of approximately 25%, were placed into molds measuring 8 cm in length, 2 cm in height, and 2 cm in width. The samples were vibrated on a vibrating table to reduce the air content within the specimens, enhancing compaction and minimizing internal defects. After molding, the resulting bars were dried at room temperature for 5 days. Subsequently, the samples were fired in an electric furnace at 950 °C for 30 minutes. After sintering, the bricks were labeled as S0RHA-950, S5RHA-950, S15RHA-950, and S30RHA-950.

The thermal behavior of the raw materials and the mixtures was studied. Simultaneous thermo gravimetric (TG) and differential thermal analysis (DTA) were performed up to 1100 °C. Both thermal analyses were carried out simultaneously at a heating rate of 10 °C/min in an air

atmosphere, using a Rigaku Evo plus II system (Japan).

The identification of crystalline phases present in the starting materials and in the sintered bricks was performed by X-ray Diffraction (XRD) using a Panalytical Empyrean 3G diffractometer with a CuK α radiation source and the Pixcel3D hybrid pixel detector with Medipix3 technology. The X-ray tube operated at 45 kV and 40 mA. Data were collected between 5° and 90° (2 θ) with the Bragg-Brentano vertical geometry ($\theta/2\theta$) in flat reflection mode.

The content of each phase was calculated using the Rietveld method with HighScore Plus software v2024. The Rietveld method is a full-pattern fitting technique applied to X-ray diffraction data. It refines a calculated diffraction pattern by iteratively adjusting parameters such as lattice parameters, atomic positions, peak shapes, and instrumental factors to achieve an optimal fit with the experimental pattern. This approach enables a precise quantification of the crystalline phases present in the sample [14, 15]. In addition, the amorphous phase content was determined using the internal standard method. This technique involves adding a known quantity of a reference material, in this case, 20 wt.% alumina to the sample [14-17]. By comparing the intensity of the diffraction peaks from the internal standard with those of the sample, the method allows for the accurate calculation of the amorphous fraction, which does not contribute distinct peaks in the diffraction pattern.

To evaluate the accuracy of the quantification of crystalline and amorphous phases, the relative error associated with each phase was calculated using the following equation:

$$\text{relative error} = \frac{\text{phase fraction}}{100} * \frac{R_{wp}}{100} \quad (A)$$

Where the phase fraction (%) corresponds to the percentage obtained from the Rietveld refinement, and the R_{wp} reflects the quality of the fit between the experimental and calculated data [14, 18]. This calculation enabled the interpretation of the reliability of the obtained values based on the fitting parameters. The method was applied to all phases identified in the bricks without RHA and those with 5%, 15%, and 30% RHA content (by weight).

The microstructural analysis was performed using a field emission scanning electron microscope (FE-SEM), with energy-dispersive X-ray spectroscopy (EDAXS) for elemental analysis (Jeol, S-4700 I, Japan).

The apparent density and open porosity of the sintered samples were determined following the Archimedes method [16]. Additionally, the relative densities (RD) of the composites were calculated as the ratio between the apparent and theoretical densities. The theoretical densities were derived considering the mineralogical composition obtained through Rietveld refinement, which provides a quantitative phase analysis of the crystalline and amorphous phases in the material. [14, 16]. This approach allows for a more accurate estimation of the theoretical density by accounting for the specific contributions of each phase to the overall material density. Total porosity (P) was calculated as:

$$P = (1 - RD) * 100 \quad (B)$$

To analyze the mechanical properties of the mixtures with different rice husk ash contents, three-point bending tests were conducted to determine the modulus of rupture (MOR) for each formulation. The tests were performed on rectangular section bars, with a support span of 56 mm and a displacement speed of 1 mm/min, using an Instron universal testing machine (model 5985).

The dynamic Young's modulus (E) for each brick was determined at room temperature using the impulse excitation technique (Grindosonic MK5 Industrial, Lemmens, Belgium) following the ASTM C1198 standard. [19, 20]. The reported dynamic Young's modulus is the average of 10 measurements, as specified in:

$$E = 0.94642 \frac{\rho l^4 f^2 T}{t^2} \quad (C)$$

where ρ is the materials density, l is the length of the bar, f is the fundamental resonant frequency, t is the thickness of the beam, and T is a correction factor provided by Spinner and Tefft [19, 21].

The measured E values were compared to the theoretical moduli at zero porosity (E_0), which were estimated based on the volumetric fraction (V_i) and the elastic modulus (E_i) of each phase present, using the phase rule. The corresponding equation is as follows:

$$E_{mix} = \sum_i^n V_i * E_i \quad (D)$$

E_{mix} = Modulus of Elasticity of the material.

V_i = Volumetric fraction of phase

i = phases

n = Number of phases present in the material.

To further analyze the behavior of the experimental elastic modulus about the microstructure and porosity of bricks with different RHA contents, the results were compared with three theoretical models for composite materials:

I. Dispersed Esferoids Differential (DEM) Model: This model considers the dispersed phase (RHA particles) embedded in the continuous phase (clay matrix), accounting for the interaction between the two phases to estimate the elastic modulus of the composite. It is noteworthy that the elastic modulus values were expressed in gigapascals (GPa)

$$E_{brick} = E_m \left(1 - \frac{V_f (E_m - E_f)}{V_f E_f + E_m (1 - V_f)} \right) \quad (E)$$

E_{brick} : Modulus of elasticity of the materials.

E_m : Modulus of elasticity of the pure matrix.

E_f : Modulus of elasticity of the dispersed phase RHA.

V_f : Volume fraction of the dispersed phase RHA, expressed as a decimal.

II. Power Law Model: This model establishes an empirical relationship between the composite's elastic modulus and the proportion of individual components,

capturing the nonlinear mechanical behavior of the material.

$$E = E_0 (1-P)^m \quad (F)$$

Where E is the modulus of elasticity without porosity, P is the porosity fraction, and m is an empirical parameter that depends on the material's microstructure; in this case, a value of $m = 3.5$ was used.

III. Hasselman-Johnson Model: This model is used to analyze the influence of dispersed phases in the matrix and how the mechanical properties and phase interactions affect the materials stiffness, while also considering the impact of porosity.

$$E = E_m (1-\alpha V_f)^n \quad (G)$$

where

E_m : is the modulus of elasticity of the pure matrix (assumed to be 10 GPa).

V_f : is the volume fraction of the dispersed phase (5%, 15%, and 30%).

α : is the empirical correction factor (assumed to be 1).

n : is the empirical exponent (assumed to be 3).

RESULTS AND DISCUSSION

The XRD diffractograms of the raw materials were analyzed using the following ASTM files: PDF 461045, for quartz SiO_2 ; PDF 000340175 for Muscovite $2\text{M}_2\text{Al}_2\text{H}_2\text{KO}_{12}\text{Si}_4$; PDF 841455 for Microcline ($\text{K}_{0.95}\text{Na}_{0.05}$) AlSi_3O_8 ; PDF 01-089-8104 for Hematite.

Figure 1a shows the X-ray diffraction (XRD) pattern corresponding to the two clays used for brick production. Quartz (SiO_2) was identified as the main phase in both samples, while the secondary phases include anorthite, albite, and muscovite. Additionally, small amounts of hematite were detected, which was confirmed by the presence of the characteristic peaks at 33.2° and 35.7° (2θ), typical of this mineral phase. These phases coincide with the characteristic

phases present in these types of soil [11,12,22].

Fig 1b shows the XRD of the RHA. The amorphous structure is more clearly observed throughout the diffractogram, which is due to the structure that the silica presents under the conditions in which the ash was obtained.

In Figure 2, the TG curve (black) represents the thermogravimetric analysis of the sample, showing an initial sharp decrease in mass around 60°C , attributed to the evaporation of absorbed moisture. This is followed by a slower, continuous reduction in mass as the temperature increases, reflecting further stages of decomposition, including the breakdown of organic material within the RHA. The total mass loss observed is approximately 4% by 1100°C .

The DTA curve (blue) illustrates the corresponding thermal events, with exothermic or endothermic reactions associated with phase transitions or decomposition events. These thermal changes, seen as peaks or troughs in the DTA curve, provide additional insight into the transformations observed in the TG curve. This combined thermal analysis offers valuable information about the behavior of RHA during the firing process, critical for optimizing its use in brick production. The correlation between the TG and DTA curves allows for a deeper understanding of the simultaneous evolution of mass loss and thermal behavior. While the transformation of amorphous silica into crystalline phases around 700°C , as reported by other authors [9, 23-26] is not observed in these curves; this may be due to differences in heating rates, sample composition, or experimental conditions. Subtle transformations like these often require more precise instrumentation or specific experimental setups to be detected clearly.

The micrograph of the rice husk ash (Figure 3a) reveals a highly porous structure, with numerous small pores uniformly distributed throughout the sample. Although it is not possible to accurately determine the average particle size from the SEM micrograph, it is observed that particles of $50\ \mu\text{m}$ present a fine and lightweight texture.

Figure 3b presents the results of the EDAXS analysis

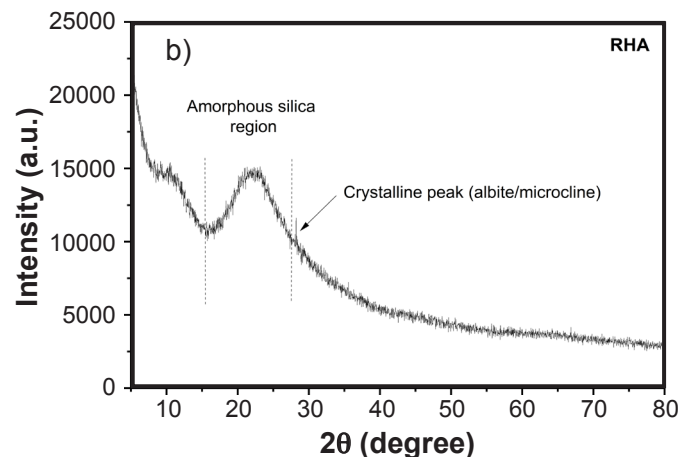
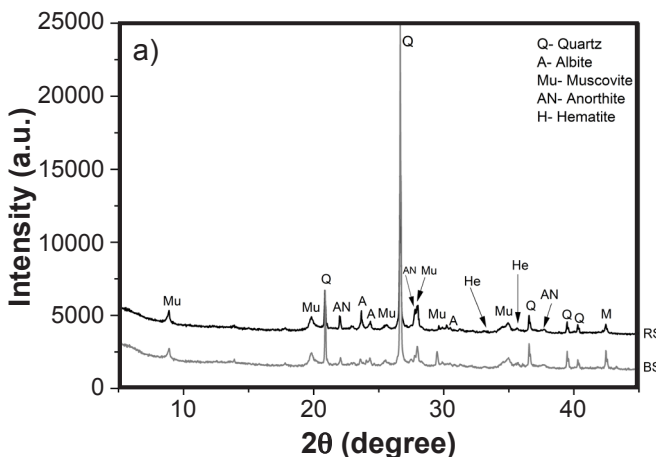


Figure 1a: XRD of the clays used to obtain the ceramic bricks. Q: Quartz, A: Albite, AN: Anorthite, Mu: Muscovite, H: Hematite.

Figure 1b: XRD patterns of RHA ash used as an additive in the manufacture of ceramic bricks

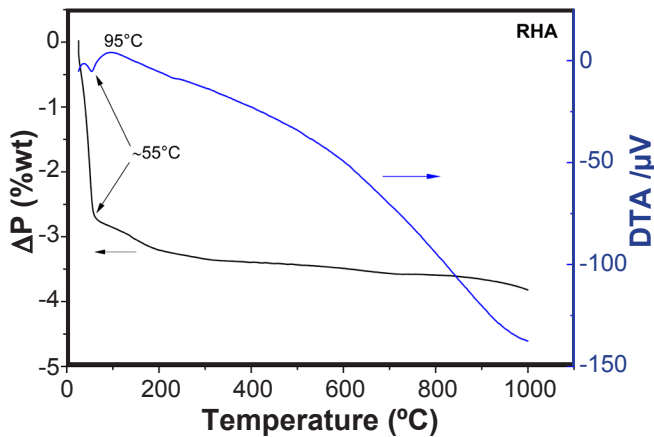


Figure 2: DTA/TG curves of the rice husk ash (RHA) used for brick manufacturing.

(Energy Dispersive X-ray Analysis), revealing a high silica content, which suggests its potential as a pozzolanic material. Additionally, smaller proportions of other elements, such as carbon, aluminum (Al), potassium (K), magnesium (Mg), and calcium (Ca), were detected, which may influence the mechanical and reactive properties of the ash in construction applications. The results obtained by EDAXS are also shown in Table 1.

The DTA/TG analyses of the various mixtures used in the production of ceramic bricks are presented in Figures 4a, 4b, 4c, and 4d, corresponding to formulations containing 0%, 5%, 15%, and 30% rice husk ash (RHA), designated as: V0RHA, V-5RHA, V-15RHA, and V-30RHA respectively. In all cases, the thermogravimetric analysis (TG) indicates a weight loss ranging between 8% and 10%.

The blue curves (TG derivative) provide clearer insights into the changes occurring in the samples during weight loss. These curves reveal that the initial thermal transformations occur around 60 °C, which is directly related to the evaporation of water contained in the clay components of the samples. Two peaks are distinguished, one at approximately

400 °C and the other near 600 °C [27]. These changes are attributed to the loss of organic material present in the clays used, as well as the decomposition of carbon contained in the rice husk ash (RHA), resulting from its incomplete combustion during the formation process. Specifically, for the samples containing RHA, the weight loss observed in the TG analysis between 400 °C and 600 °C can be directly correlated to the nature of the ash. Previous studies have shown that this type of ash exhibits significant weight loss in this temperature range due to the decomposition of its remaining carbonaceous components. The incomplete combustion of the husk during the generation of the ash leaves carbon residues that oxidize within this interval, contributing to the weight reduction of the samples [28].

It is worth mentioning that the small variations in the dehydration temperature observed between 60 °C and 66 °C in the different samples may be due to several factors, such as the composition and proportions of the materials used, the initial water content of each sample, and the presence of RHA [25]. Differences in the amount of water adsorbed in the clays or how it is distributed in the mixture can cause evaporation to begin at slightly different temperatures. Furthermore, the presence of rice husk ash (RHA) can alter the thermal properties of the sample, as RHA may contain volatile components that influence the thermal behavior of the mixture.

Figure 5 presents the X-ray diffraction (XRD) patterns of the bricks, both without rice husk ash (RHA) and with 5%, 15%, and 30% RHA, fired at 950 °C for 30 minutes. These patterns reveal the presence of several crystalline phases. The primary phase identified in all compositions is quartz (SiO_2) [PDF card 03-065-0466], which is expected in traditional clay-based bricks due to the high silica content of the raw materials [23]. Quartz plays a crucial role in the mechanical stability of the bricks, providing a rigid and durable framework.

Additionally, the XRD analysis identified secondary crystalline phases, including muscovite [PDF card

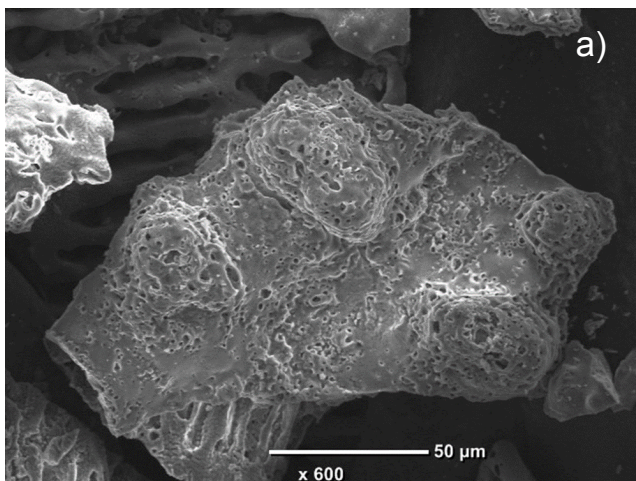


Figure 3a: Grain of RHA taken by SEM x 600

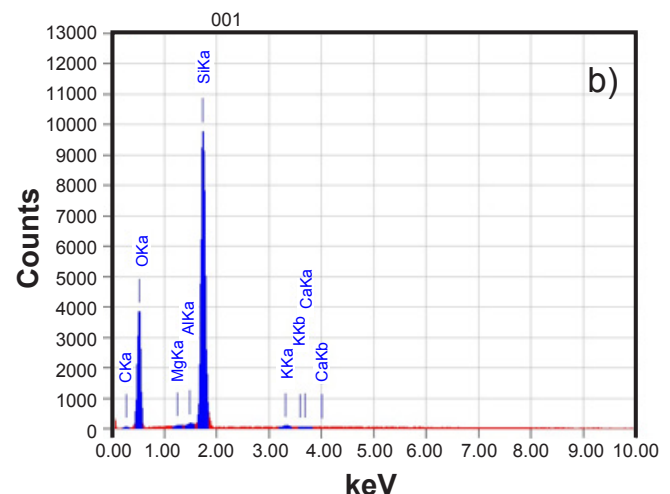


Fig 3b: EDAXS Spectrum of Rice Husk Ash (RHA): Identification of Main Chemical Components.

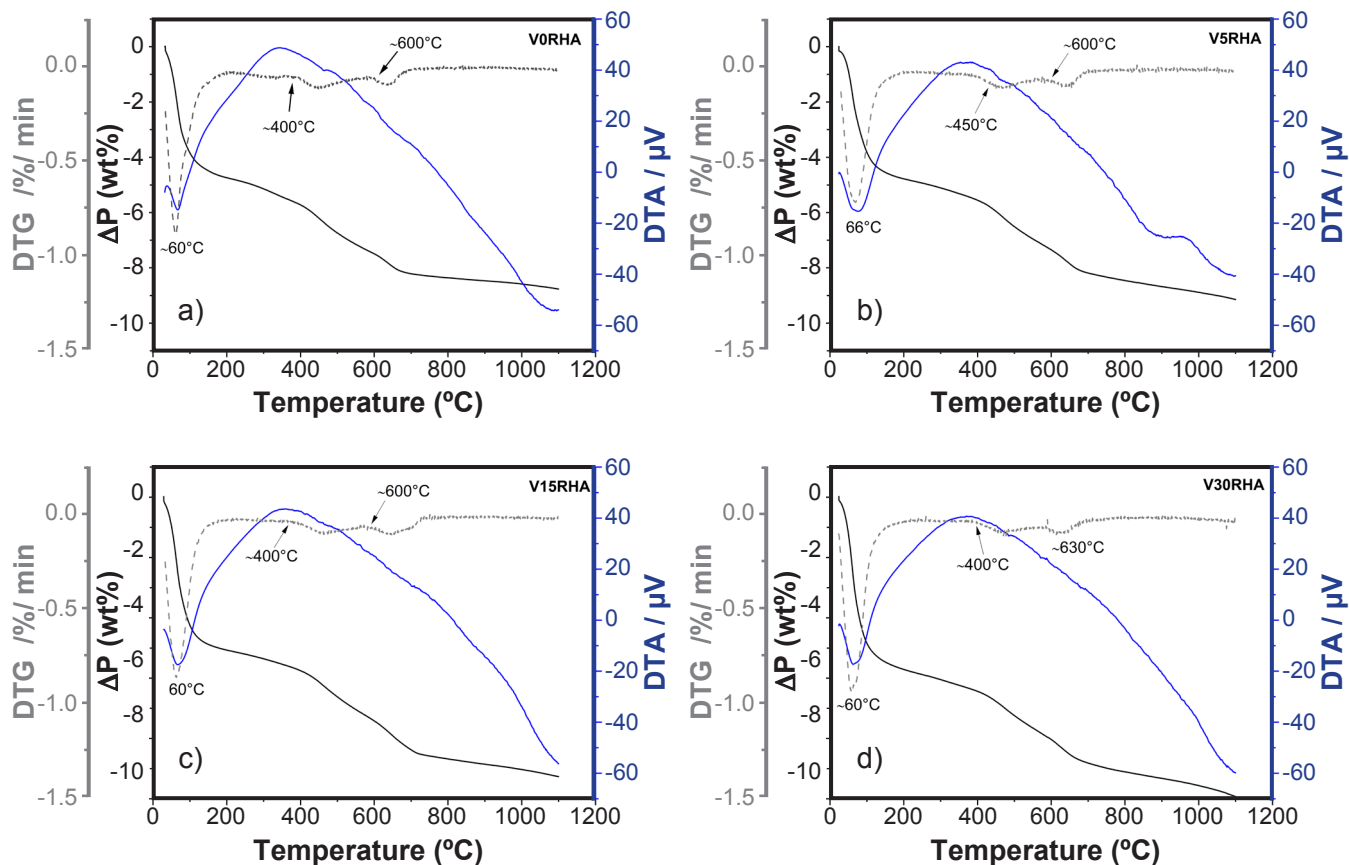


Figure 4: a, b, c, and d: ATD/TG of the mixtures V0RHA, V-5RHA, V-15RHA, and V-30RHA respectively.

Table 1 - Chemical analysis of the oxides present in rice husk ash obtained by EDAXS.

Components	Weight (%)
C	8.39
SiO ₂	90.36
Al ₂ O ₃	0.34
K ₂ O	0.56
MgO	0.16
CaO	0.16

00-034-0175], albite (plagioclase feldspar) [PDF card 01-089-6429], microcline (potassium feldspar) [PDF card 01-076-0831], and hematite [PDF card 01-089-8104]. These phases contribute differently to the microstructure and potential material properties. Albite and microcline are commonly associated with fluxing behavior, which can facilitate densification and vitrification at sufficiently high temperatures. However, in this case, their low concentration suggests that their influence on sintering is limited. Instead, the presence of RHA influences the brick's microstructure more significantly, primarily through its impact on porosity development[29].

The presence of muscovite, though minimal, suggests the persistence of phyllosilicates in the ceramic matrix. Its thermal stability at these firing conditions indicates that it does not undergo significant decomposition, and while its

low thermal conductivity could provide some insulating effects, its small quantity makes this effect negligible. Hematite, present in trace amounts, is primarily responsible for the reddish coloration of the fired bricks.

The same secondary crystalline phases identified in bricks without RHA are present in bricks containing RHA, and their relative proportions do not exhibit significant variations with increasing RHA content. Similar trends have been reported in studies where agro-industrial ashes, depending on their

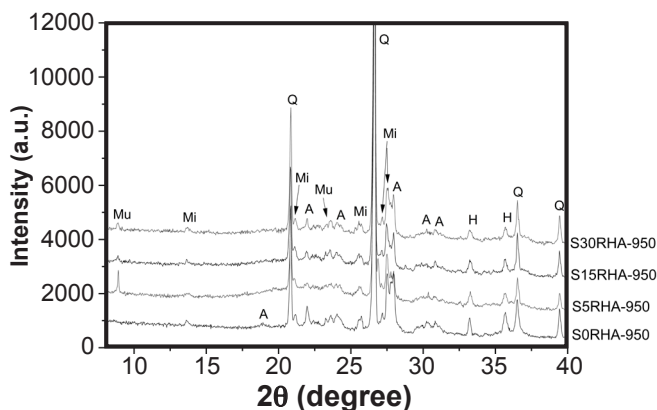


Figure 5: XRD of the different bricks sintered at 950°C. Q: Quartz (SiO₂), Mu: Muscovite (K,Na)Al₂(Si,Al)₄O₁₀(OH)₂, A: Albite, Na(AlSi₃O₈), Mi: Microcline (K_{0.94}Na_{0.06}Al_{1.01}Si_{2.99}O₈), H: Hematite (Fe₂O₃).

Table 2 - Results of the Rietveld refinement for S5RHA-950, S15RHA-950, S30RHA-950, and S0RHA-950.

Name	Crystalline phase content (wt.%)						Rwp
	Quartz	Albite	Microcline	Hematite	Muscovite	Amorphous	
S5RHA-950	22.3	10.1	4.7	1.9	2.3	58.7	5.94
S15RHA-950	24.7	10.3	5.2	1.7	1.8	56.3	4.77
S30RHA-950	27.8	11.5	6.4	2.1	1.9	50.3	4.49
S0RHA-950	32.3	30.3	8.7	2	0	26.7	4.7

composition and processing temperature, influence phase formation and microstructure development [24].

The content of each crystalline and amorphous phase is presented in Table 2, alongside the corresponding Rwp values from Rietveld refinement, which quantify the quality of the fit between experimental and calculated data. The Rwp values for S5RHA-950, S15RHA-950, and S30RHA-950 were 4.77%, 4.49%, and 4.70%, respectively, while the brick without RHA (S0RHA-950) exhibited a slightly higher Rwp value of 5.94%. Lower Rwp values indicate a better fit, suggesting that the refinement adequately represents the crystalline structure of the samples. The Goodness of Fit (GOF), calculated as the ratio of Rwp to the expected R-value (R_{exp}), further confirms the reliability of the refinements [18]. GOF values close to 1 were observed for samples with RHA (1.47 for S5RHA-950, 1.42 for S15RHA-950, and 1.51 for S30RHA-950), indicating high-quality fits.

The relative error for each phase was calculated as described in the Materials and Methods section. For example, the quartz phase exhibited a relative error of 1.27% in the S5RHA-950 sample, 1.09% in S15RHA-950, and 1.92% in S30RHA-950, reflecting a slight increase with higher RHA content. The amorphous phase, which represents a larger fraction compared to crystalline phases, displayed higher relative errors ranging from 2.34% to 1.59%. Minor phases, such as hematite, showed relative errors below 0.12%, consistent with their smaller quantities. These results, combined with the Rwp and GOF values, provide a solid basis for interpreting the phase composition and the effects of RHA addition on the material properties.

The analysis of bricks with rice husk ash (RHA) shows that porosity progressively increases with the addition of this material due to various factors occurring during the firing process. The results presented in Table 3, obtained using the Archimedes method and the phase rule, confirm this trend and demonstrate how thermal mechanisms influence porosity behavior. The key factors contributing to this phenomenon are detailed below:

a. Presence of organic material in RHA: RHA contains organic matter that combusts at high temperatures (950 °C), leaving voids or pores in the brick structure. As the RHA content increases, a larger amount of organic material is burned off, creating additional voids and increasing overall porosity [30, 31].

b. Particulate nature of RHA: RHA consists of fine, lightweight particles that do not compact effectively with clay minerals during firing. This leads to the formation of

voids and an increase in porosity [32, 33]. As more RHA is added, achieving a dense and compact structure becomes increasingly difficult, resulting in a higher number of pores.

c. Silica reactions in RHA: Although the silica present in RHA could interact with clay minerals, promoting the crystallization of phases such as quartz and albite, these reactions may not fully compensate for the increased porosity caused by the combustion of organic material and the lightweight nature of RHA [34-36]. As the RHA content increases, a decrease in the amorphous phase is observed, suggesting that the silica in RHA might favor the crystallization of existing phases rather than forming new compounds. This crystallization could explain the reduction in the amorphous phase in materials with higher RHA content, although this effect requires further evaluation to be confirmed.

d. Limited sintering of RHA: Unlike pure clay, RHA does not facilitate effective sintering, which limits pore closure and leads to an increase in the number of open pores in the brick structure [37].

Despite the significant increase in porosity with higher RHA contents (from 28.02% to 46.0%), the theoretical density (D_{Th}) remains relatively constant, with values ranging from 2.56 to 2.59 g/cm³ [38]. This suggests that crystalline phases do not undergo significant changes. However, the apparent density (D_a) slightly decreases, from 1.65 g/cm³ to 1.60 g/cm³, reflecting the formation of additional pores. This slight reduction in D_a, combined with the substantial decrease in the amorphous phase (Table 3), could be explained if RHA were inducing crystallization in certain phases, which might partially compensate for the effects of increased porosity. However, this hypothesis requires further analysis to determine the actual impact of RHA on microstructure evolution.

The relative error for both apparent density (D_a) and porosity was calculated, providing a measure of uncertainty about the experimental values. Despite these uncertainties, the data indicate that although higher porosity reduces apparent density, the possible crystallization of new phases could stabilize theoretical density, counteracting, to some extent, the impact of increasing RHA content. Table 3 presents the results for porosity (P), apparent density (D_a), and theoretical density (D_{Th}).

The experimental results show a significant reduction in the modulus of rupture (MOR), decreasing from 8.5 MPa for 5% RHA to 7 MPa for 15%, and down to 5 MPa for 30% RHA. This reduction is primarily attributed to the increase in porosity, which introduces stress concentrators

Table 3 - Results of porosity (P), apparent density (Da), and theoretical density (Dth).

Name	Da (g/cm ³)	DTh (g/cm ³)	Porosity (%)
S0RHA950	1.65 (±0,05)	2.57	28.0 (±0.6)
S5RHA950	1.65 (±0,06)	2.59	29.70 (±0.09)
S15RHA950	1.64 (±0,1)	2.57	37.75 (±0.07)
S30RHA950	1.6 0(±0,08)	2.56	46.0 (±0.1)

in the form of voids and microcracks, weakening the material under load [32, 39]. The decrease in MOR could be explained if it is assumed that the RHA particles are larger and irregular in shape, which would account for the formation of microcracks during the sintering process, making the material more susceptible to stress accumulation [32]. The combustion of organic matter in the RHA generates gas bubbles during sintering, which become trapped in the matrix, further contributing to the development of pores [30, 36]. Incomplete densification due to suboptimal sintering conditions, such as lower temperatures or insufficient sintering time, also plays a significant role in the final mechanical properties [32]. These factors, among others, lead to a reduction in the material's mechanical strength, resulting in an overall decrease in strength.

Additionally, the observed variability in the MOR data includes an estimated margin of uncertainty between 5% and 7%, which can be attributed to the intrinsic heterogeneity of the material. The uneven distribution of pores and microcracks across the samples, as well as possible inconsistencies in sample preparation, contribute to this variability. The firing conditions, including temperature, heating rate, and sintering time, also affect the results by influencing the degree of densification and crystallization. These factors slightly impact the repeatability of the results, generating a margin of error in the obtained values. This behavior is consistent with the data presented in Table 3, where the porosity increases from 28.02% for S0RHA-950 to 46% for S30RHA-950. The increase in porosity correlates with the reduction in flexural strength observed in the three-point bending tests, where the modulus of rupture decreases from 8.5 MPa for 5% RHA to 5 MPa for S30RHA-950. The increased formation of pores weakens the material's structural integrity by acting as stress concentrators.

In comparison, conventional fired clay bricks typically have an average MOR of 4.0 MPa. Despite the decrease in MOR with higher RHA content, bricks containing low percentages of RHA (5%) remain competitive and, in some cases, superior to traditional bricks [39]. This demonstrates that, although the mechanical properties diminish with higher RHA content, the material retains adequate performance for certain applications.

Table 4 presents the density and elastic modulus values for each phase present in the bricks, obtained from the literature. These values were used to calculate the density and elastic modulus of the materials, applying the phase rule.

Table 4 - Theoretical Density and Elastic Modulus of Phases Present in the Material.

Density and Modulus of Elasticity of the Present Phases.		
Present Phases	Theoretical density (g/cm ³)	E (GPa)
Quarz	2.65	70 [41]
Albite	2.62	75 [41]
Muscovite	2.83	60 [42]
Microcline	2.56	50 [42]
Hematite	5.25	210 [42]
Amorphous phase	2.50	70 [41,42]

The experimental elastic modulus for different samples with varying percentages of rice husk ash (RHA) is shown in Figure 6. To better understand the mechanical behavior of the material, several predictive models were applied to compare how each model reflects the influence of porosity on the elastic modulus [43, 44]. These models include the Dispersed Esferoides Differential (DEM) model, the Power Law model, and the Hasselman-Johnson model, all of which take different approaches to describe the impact of porosity on mechanical properties.

The DEM model assumes the presence of isolated spherical pores, which may not fully represent the interconnected porosity observed in the experimental samples. In contrast, the Power Law model assumes a more distributed porosity, making it more suitable for materials with a network of interconnected voids, such as those observed in bricks with high RHA content. Finally, the Hasselman-Johnson model accounts for pore connectivity and its effect on mechanical behavior, providing a more comprehensive understanding of the material's response to stress.

Table 5 presents the elastic modulus (E) values calculated using the phase rule, assuming zero porosity, for bricks with different percentages of RHA. The values obtained for samples with 5%, 15%, and 30% RHA are 72.76 GPa, 71.74 GPa, and 72.07 GPa, respectively. Despite the differences in RHA content, the calculated elastic modulus values remain very similar, indicating consistency in the elastic behavior of the bricks when porosity is not considered. However, it is essential to note that these values represent an ideal condition with no porosity, which differs from the values obtained under experimental conditions.

Figure 6 shows a comparison between the experimental

Table 5 - Values of the modulus of elasticity calculated from the rule of phases eq (B)

Name	E (GPa)
S5RHA-950	72.76
S15RHA-950	71.74
S30RHA-950	72.07

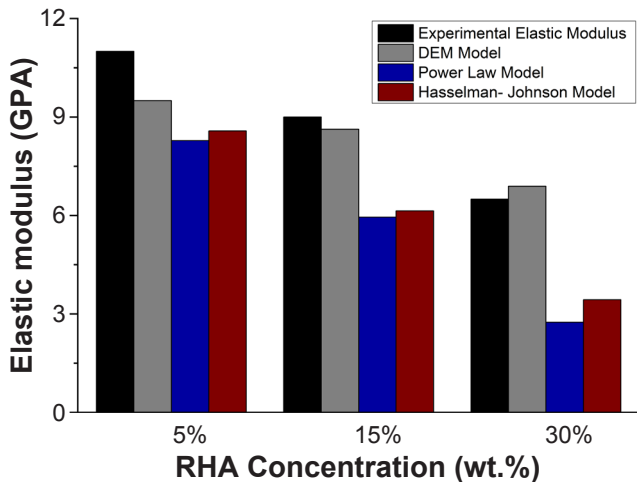


Figure 6: Comparison of Elastic Modulus for Bricks with various RHA concentrations: Experimental Data vs. Predicted Models (DEM, Power Law, Hasselman-Johnson)

values and the predicted values from the three theoretical models of the elastic modulus for bricks with different weight percentages of RHA, allowing an evaluation of how well each model fits the experimental data. The x-axis represents the weight percentages of RHA used (5%, 15%, and 30%), while the y-axis represents the elastic modulus in GPa.

The dark gray bars represent the experimental values of the elastic modulus measured in the bricks. The light gray bars correspond to the values predicted by the DEM model, which provides the best fit to the experimental data.

The dark blue bars represent the values predicted by the Power Law model, which significantly overestimates the elastic modulus, particularly at higher RHA concentrations. On the other hand, the crimson bars correspond to the values predicted by the Hasselman-Johnson model, which also overestimates the elastic modulus for all RHA concentrations.

A decreasing trend in the elastic modulus is observed as the RHA content in the bricks increases. This reduction is accurately captured by the DEM model, suggesting that this model is more suitable for describing materials with high porosity and interconnected pores. In contrast, the Power Law and Hasselman-Johnson models fail to adequately capture the magnitude of the reduction in the elastic modulus, likely due to their assumption of a more uniform pore distribution.

By comparing the experimental data with the predictions of these models, it is clear that the DEM model best fits the actual behavior of the material. Its ability to account for the high level of porosity and the interconnected nature of the pores makes it the most reliable tool for predicting the elastic modulus in materials with high RHA content [43,44].

Furthermore, when calculating the relative errors between the experimental and predicted values, it was found that the DEM model exhibited the smallest relative errors across all RHA concentrations tested.

For the samples with 5% RHA, the DEM model

showed a relative error of 5.88%, while the Power Law and Hasselman-Johnson models showed errors of 5.88% and 8.24%, respectively.

As the RHA content increased, the DEM model continued to show the best fit, with relative errors of 1.43% and 4.00% for 15% and 30% RHA, respectively.

In contrast, the other models showed a considerable increase in relative errors, particularly for 30% RHA, where the errors for the Power Law and Hasselman-Johnson models were 44% and 50%, respectively.

CONCLUSIONS

This study demonstrates that the incorporation of rice husk ash (RHA) into ceramic bricks can be a viable and sustainable alternative for the construction industry.

Although the increase in RHA content leads to a reduction in mechanical properties due to increased porosity, experimental results show that up to 15% RHA can be used without compromising the material's performance for non-structural applications.

The analysis of textural and mechanical properties, supported by predictive models such as the DEM model, suggests that the increase in porosity and the reduction in the amorphous phase are inherent effects of RHA usage, without the formation of new crystalline phases. This reaffirms that RHA, in the studied proportions, acts primarily as a pore-forming agent, affecting the structural cohesion of the material.

Although it was not experimentally determined that the incorporation of RHA would improve the thermal and acoustic properties of the bricks, this could represent an additional advantage in terms of energy efficiency and comfort in buildings.

This approach highlights the potential of RHA to be used in various agricultural regions, offering a solution that promotes the circular economy and efficient resource use.

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The analysis of the mechanical property values about porosity, using various theoretical models, provided a deeper insight into the relationship between porosity and the mechanical properties of the bricks. The DEM model proved to be the most accurate in predicting how porosity affects mechanical properties, particularly elastic modulus and modulus of rupture. The predictive models of mechanical

properties allow for a better understanding of the material behavior and provide useful tools for the design of ceramic bricks with RHA content, optimizing their performance based on structural needs.

Although it was not experimentally determined that the incorporation of RHA would improve the thermal and acoustic properties of the bricks, this could represent an additional advantage in terms of energy efficiency and comfort in buildings.

This approach highlights the potential of RHA to be used in various agricultural regions, offering a solution that promotes the circular economy and efficient resource use.

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